

Computer Science & Data Science Artificial Intelligence

DPP 3 ; Top 250 Questions of DBMS Part-3



DBMS : DPP 3

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[NAT]



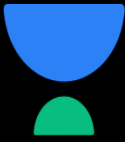
Consider relation $R(P, Q, R, S, T, U)$ with following functional dependencies:

(a) $P \rightarrow Q$ (b) $RS \rightarrow T$

(c) $T \rightarrow P$ (d) $Q \rightarrow S$

How many candidate keys does R have?_____.

[MCQ]



If a relation $R(A, B, C, D, E)$ holds the following dependencies:

$AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow A, D \rightarrow B.$

Then how many candidate keys can be made out of it _____.



2



3



1



4

[MCQ]



Consider the set of functional dependencies for a relation

$R(D, N, C, S) \{D \rightarrow N, D \rightarrow C, D \rightarrow S, C \rightarrow S\}$

Then choose the correct statement regarding the above set.

- A. $\{D\}$ is the superkey for the relation.
- B. $\{DN\}$ is the candidate key for the relation.
- C. $\{DC\}$ is the candidate key for the relation.
- D. $\{CN\}$ is the superkey for the relation.

[MCQ]



Consider a relation $X(PQR)$ with the set of FD's $[P \rightarrow QR, Q \rightarrow R, P \rightarrow Q, PQ \rightarrow R]$
What is number of FD present in minimal [canonical] cover?

- A 1
- B 2
- C 3
- D 4

[MSQ]

For all given set of FD, find the primary key from the options below, for relation R (A, B, C, D, E, F)

$\{A \rightarrow D, C \rightarrow BDE, B \rightarrow F, B \rightarrow C\}$

- A. AC could be the primary key.
- B. There are two candidate keys AC and AB.
- C. BC is the primary key.
- D. No primary key exists for the relation.



[MSQ]



Consider the relation $R(P, Q, R, S, T, U, V, W)$ with the following set of functional dependencies:

$F = \{P \rightarrow QRS, P \rightarrow T, TUV \rightarrow W \text{ and } U \rightarrow VW\}$

Which one of the FD in the F is redundant?

- A. $P \rightarrow QRS$
- B. $PS \rightarrow T$
- C. $TUV \rightarrow W$
- D. $U \rightarrow VW$



Suppose that we decompose the schema $r(A, B, C, D, E)$ into

$r_1(A, B, C)$

$r_2(A, D, E)$

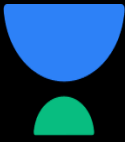
Show that this decomposition is a lossless decomposition if the following set F of functional dependencies holds: (By CHASE Test)

$A \rightarrow BC$

$CD \rightarrow E$

$B \rightarrow D$

$E \rightarrow A$



If $D_1, D_2, \dots, D_n$ are domains in a relational model, then the relation is a table, which is a subset of

A

$$D_1 \oplus D_2 \oplus \dots \oplus D_n$$

B

$$D_1 \times D_2 \times \dots \times D_n$$

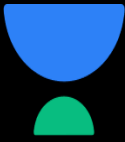
C

$$D_1 \cup D_2 \cup \dots \cup D_n$$

D

$$D_1 \cap D_2 \cap \dots \cap D_n$$

[NAT]

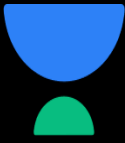


Consider the relation $R(ABCDEFGHIIKKLM)$ with the following functional dependencies set:

$\{A \rightarrow E, A \rightarrow BCD, EFG \rightarrow H, F \rightarrow GH, I \rightarrow J, K \rightarrow LM\}$

The number of redundant FD is_____.

Video Solution of DPP 3



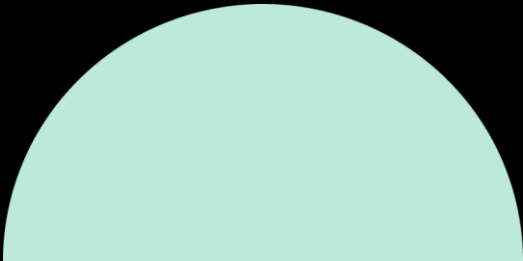
Link: <https://unacademy.com/a/Top-250-Questions-on-DBMS.html%27>

Video Solution of DPP 3 LINK



THANK YOU!

Here's to a cracking journey ahead!

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